Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

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Introduction

Online bookstores must decide which books to stock, how to price them and how to present them to customers. A publishing analytics team has asked for help understanding which factors influence the price and the star rating of books. This report documents a complete data science workflow carried out on Books to Scrape (books.toscrape.com), a practice website built for learning web scraping (Books to Scrape, 2026).

The project has five stages: collecting data with web scraping, cleaning the data, exploring it with visualisations, building a Linear Regression model to estimate book price, and reporting the findings. The full code is in the accompanying Jupyter Notebook (Kluyver et al., 2016) and the final dataset is supplied as clean_books.csv.

The main objectives are:

- to collect the title, price, star rating and stock availability of at least 200 books;
- to turn the raw data into a clean, analysis-ready dataset;
- to find patterns in prices, ratings and genres; and
- to test how well simple book information can predict price.

GIT HUB LINK

https://github.com/kingmonty202-gif/Data-Science-500-Assignment.git

Methodology

1. Data Acquisition

The data was collected using Python with the Requests library and BeautifulSoup (Richardson, 2007). The scraper loops through all 50 catalogue pages and reads each book's title, price, star rating and availability from the page HTML. It then opens every book's own page to collect the category (genre), the exact stock quantity and the description. A short pause was added between requests so that the website was not overloaded. In total, 1 000 books were collected, which is well above the minimum of 200 books required.

2. Data Cleaning

The raw data contained text where numbers were needed and a small number of missing values. The following cleaning decisions were made:

- Price: the pound sign was removed and the value converted to a decimal number so that calculations are possible.
- Star rating: the words "One" to "Five" were converted to the numbers 1 to 5, because models work with numbers.
- Stock: an in-stock indicator (1 or 0) and a quantity column were created from text such as "In stock (22 available)".
- Missing values: only 2 books had no description. They were kept, and the gap was filled with "No description", so that no price information was lost.
- Duplicates: the duplicate check found 0 duplicate rows, so no books had to be removed.
- Text fields: descriptions and titles were trimmed and converted to lower case, and the lengths of the title and description were calculated as simple numeric features.

The cleaned dataset has 1 000 rows and 12 columns, with no missing values (McKinney, 2010).

3. Visualization

Charts were created with Matplotlib (Hunter, 2007) and pandas. They include a price histogram, a count of books per rating, a box plot of price by rating, bar charts of genre size and average genre price, scatter plots, and a correlation heatmap.

4. Predictive modelling

A Linear Regression model from scikit-learn (Pedregosa et al., 2011) was used to predict price. The features were the star rating, stock quantity, title length, description length and genre. Genre was converted into 0/1 columns, using the ten largest genres and grouping the rest as "Other". The data was split randomly into 800 books (80%) for training and 200 books (20%)

for testing. The model was judged using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R-squared and compared with a baseline that always predicts the average price.

Significant findings from the Catalogue

1.1 The price landscape

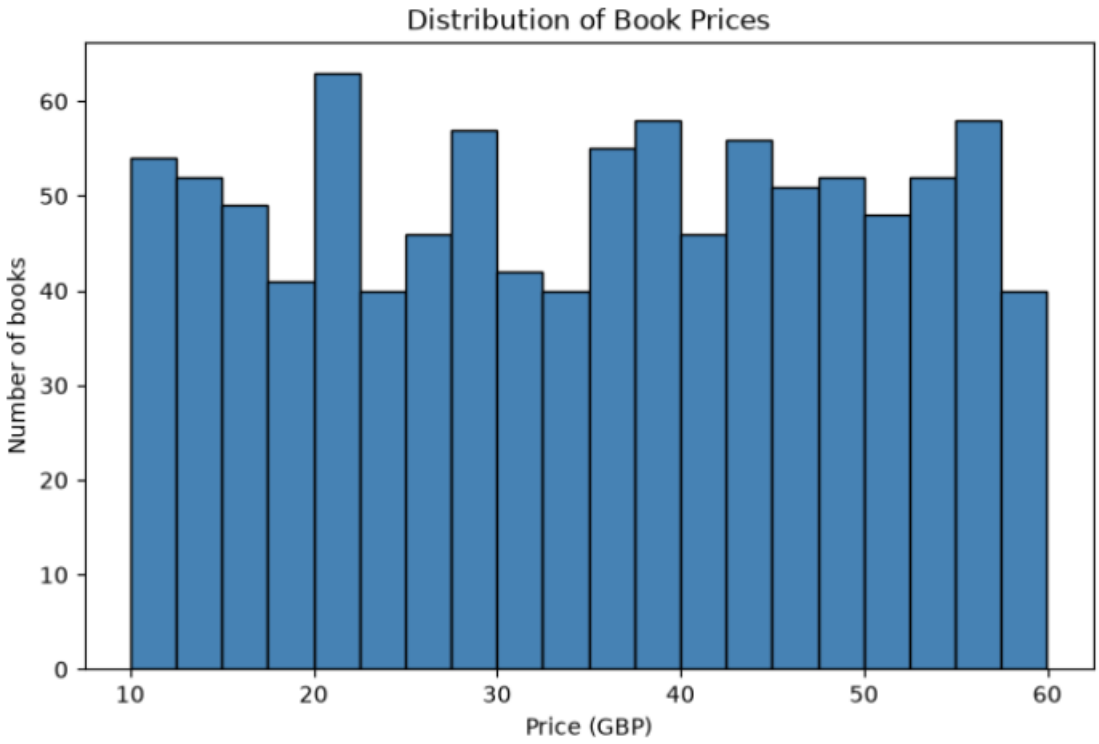


Figure 1: Distribution of book Prices

Book prices range from £10.00 to £59.99, with a mean of £35.07, a median of £35.98 and a standard deviation of £14.45. Figure 1 shows that books are spread evenly across the whole price range, with no single price band dominating and no strong skew or extreme outliers. This is unlike a real bookstore, where most prices usually cluster in a narrow range.

1.2 Star Ratings and Price

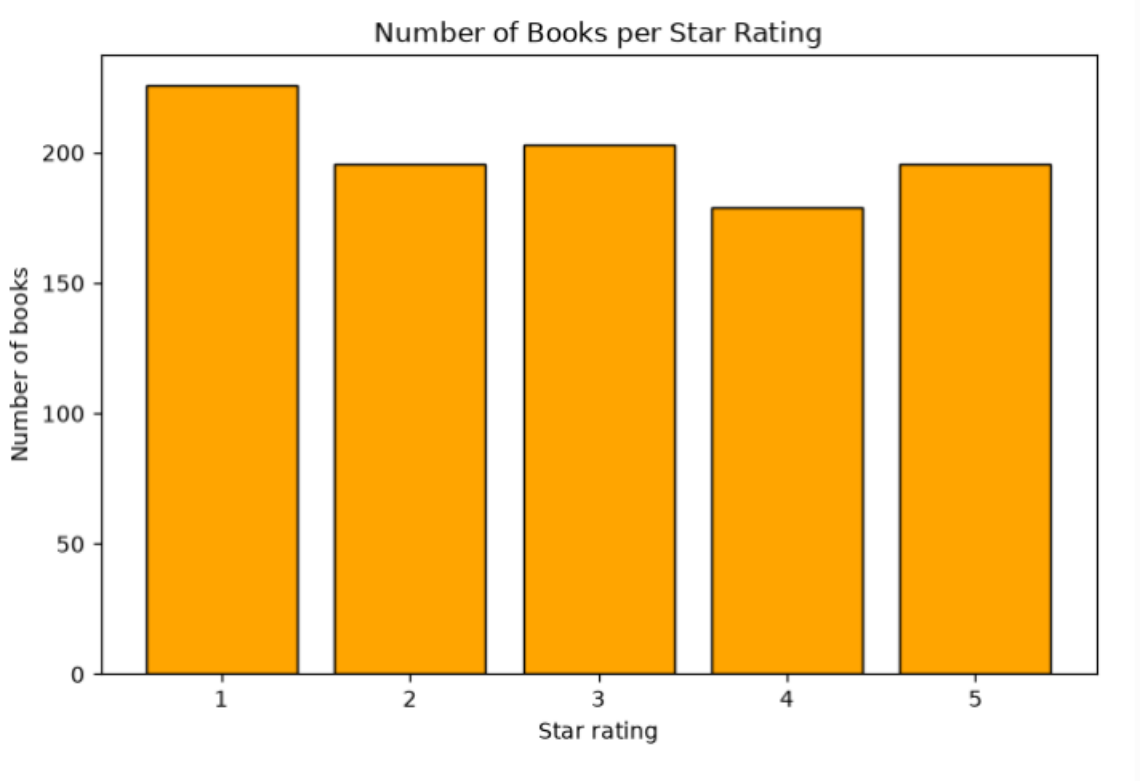


Figure 2: Number of books per star rating

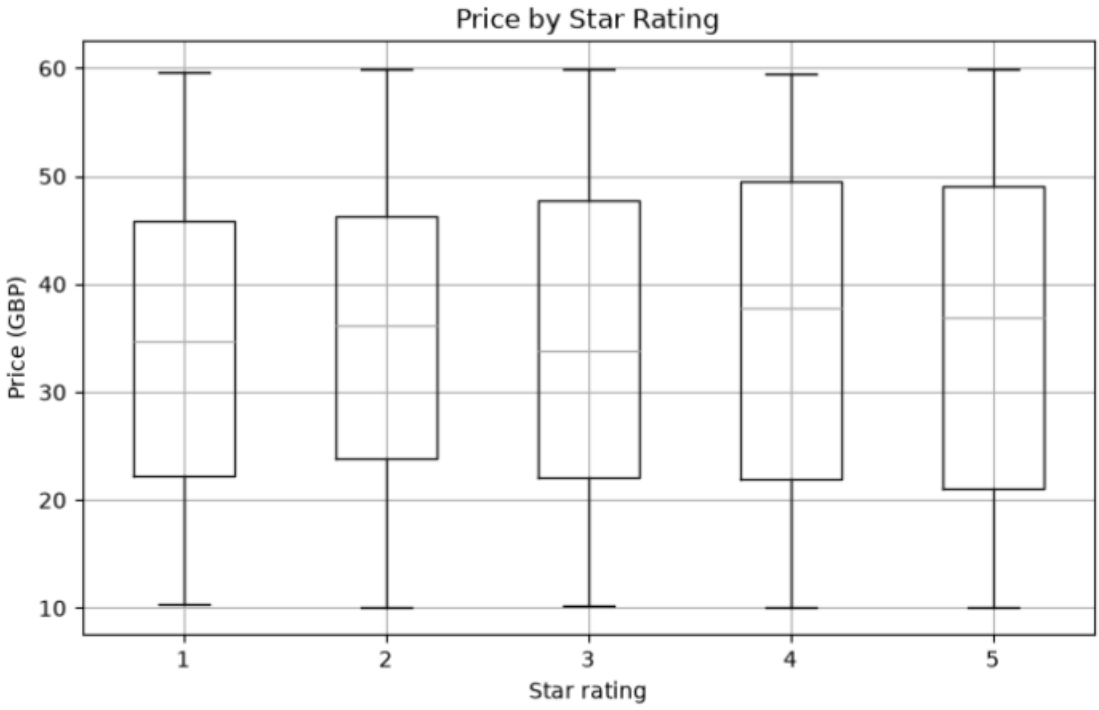


Figure 3: Price by star rating

The catalogue is split quite evenly across ratings, with one star the most common (226 books), followed by three stars (203), two and five stars (196 each) and four stars (179). The average price is £34.56 for one-star books, £34.81 for two stars, £34.69 for three stars, £36.09 for four stars and £35.37 for five stars. The gap between the cheapest and most expensive rating group is only about £1.53, so higher-rated books do not cost noticeably more.

1.3 Genre Trends

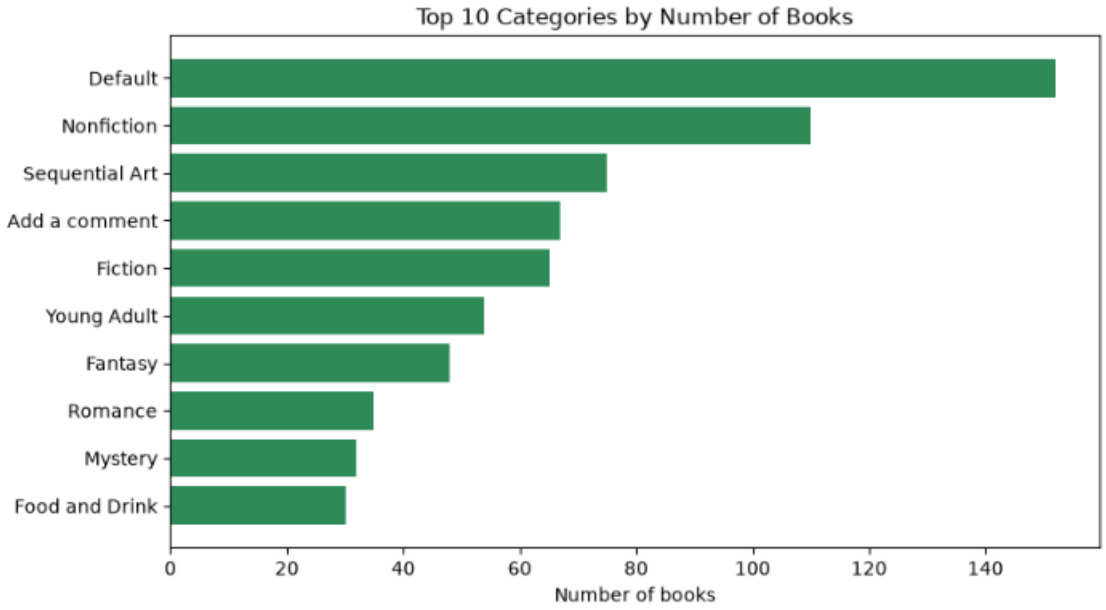


Figure 4: Top 10 categories by number of books

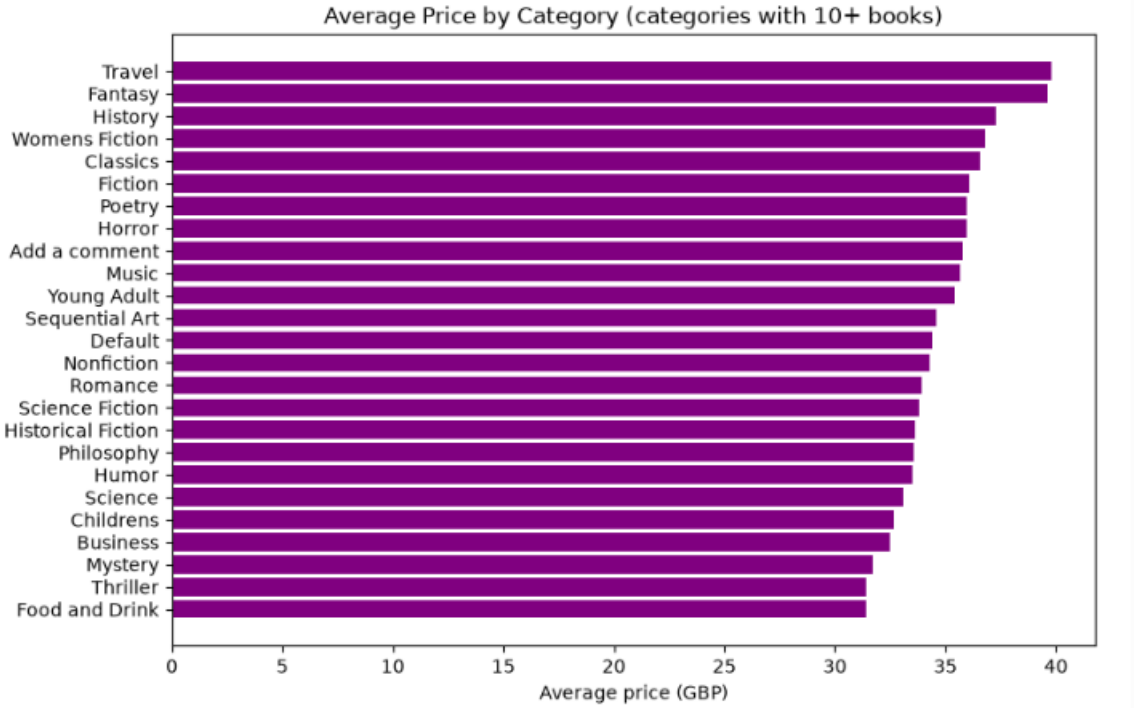


Figure 5: Average price by category (10 or more books)

1.4 Relationships between variables

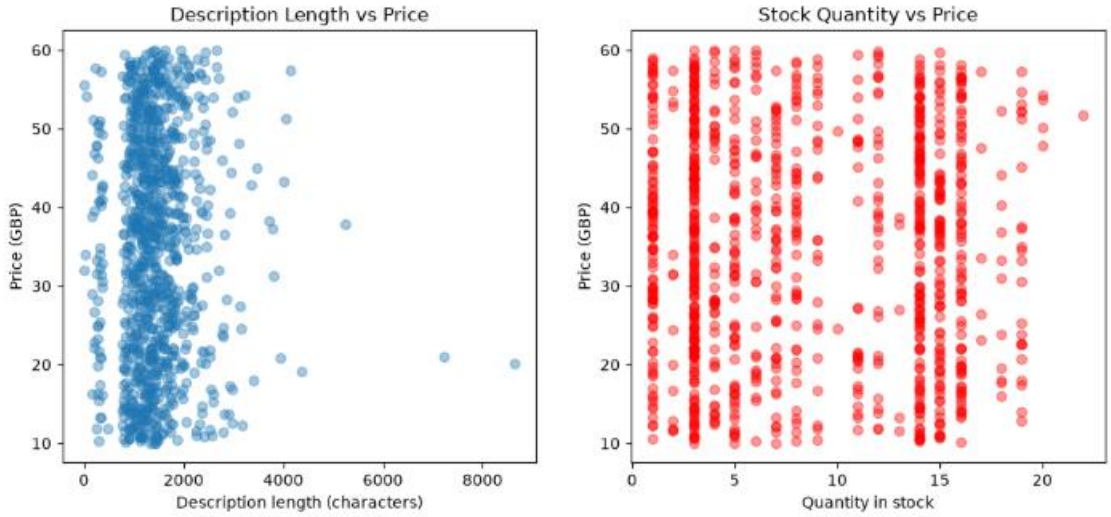


Figure 6: Description length and stock quantity compared with price

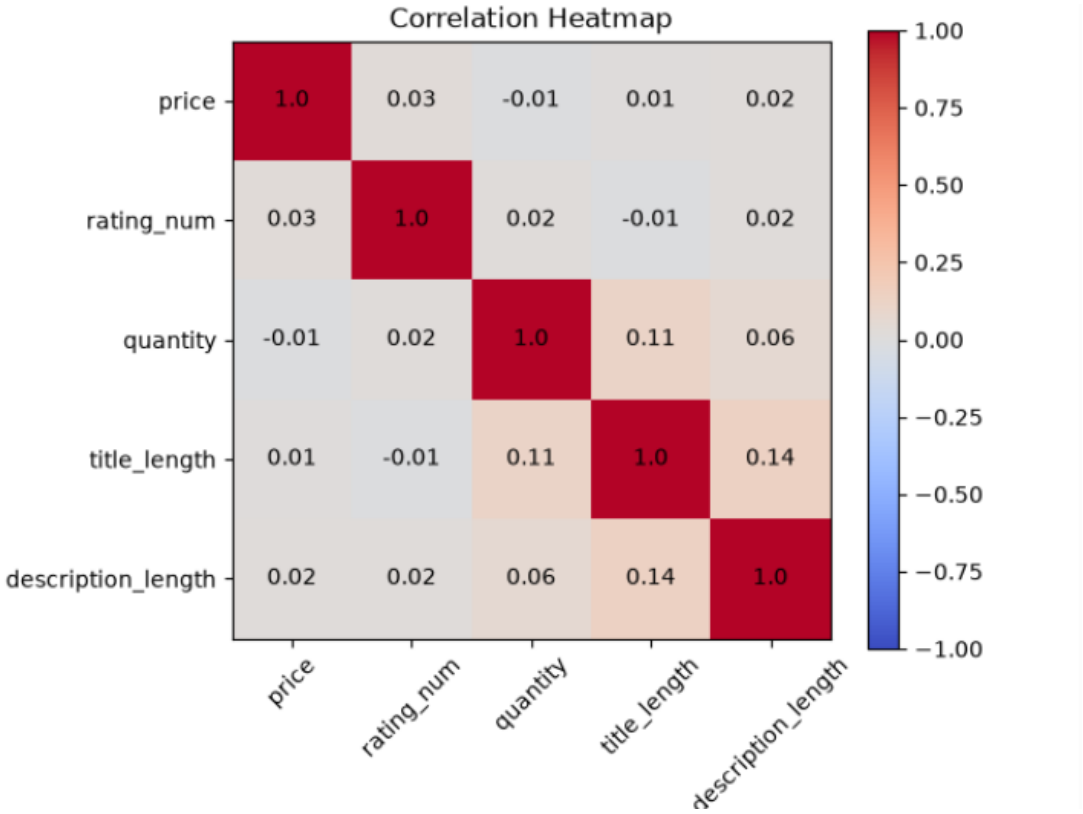


Figure 7: Correlation heatmap of the numeric variables

The correlations with price are all very close to zero: 0.03 for star rating, -0.01 for stock quantity, 0.01 for title length and 0.02 for description length. The scatter plots in Figure 6 look like random clouds of dots, with no visible pattern. The strongest relationship in the data is a weak link between title length and description length (0.14), which says nothing about price. Every book in the catalogue was in stock, so the in-stock indicator has no variation and could not help explaining price. Stock quantity ranges from 1 to 22 copies, with an average of about 8.6.

Modelling Outcomes

Table 1 shows how the Linear Regression model performed on the 200 books it had not seen during training.

Table 1: Model performance on the test set

Metric	Linear Regression	Baseline (average price)
MAE (£)	12.69	12.66
RMSE (£)	14.65	-
R-squared	-0.022	0

The model's R-squared is -0.022, which means it explains none of the variation in price and is slightly worse than simply guessing the average price for every book. Its average error (MAE) of £12.69 is almost identical to the baseline's £12.66, and its RMSE of £14.65 is about the same as the standard deviation of price itself (£14.45). In other words, the model adds no useful predictive power.

The coefficients are shown relative to the reference genre "Add a comment", which was left out when the genre columns were created. The largest effects were Mystery (-£6.88), Food and Drink (-£4.51), Nonfiction (-£3.83) and Fantasy (+£2.48). Each extra star rating changed the predicted price by only +£0.32, and stock quantity, title length and description length had effects of a few pence or less. These coefficients are small and unreliable because the model fits no better than chance, so they should be treated as descriptions of this sample only.

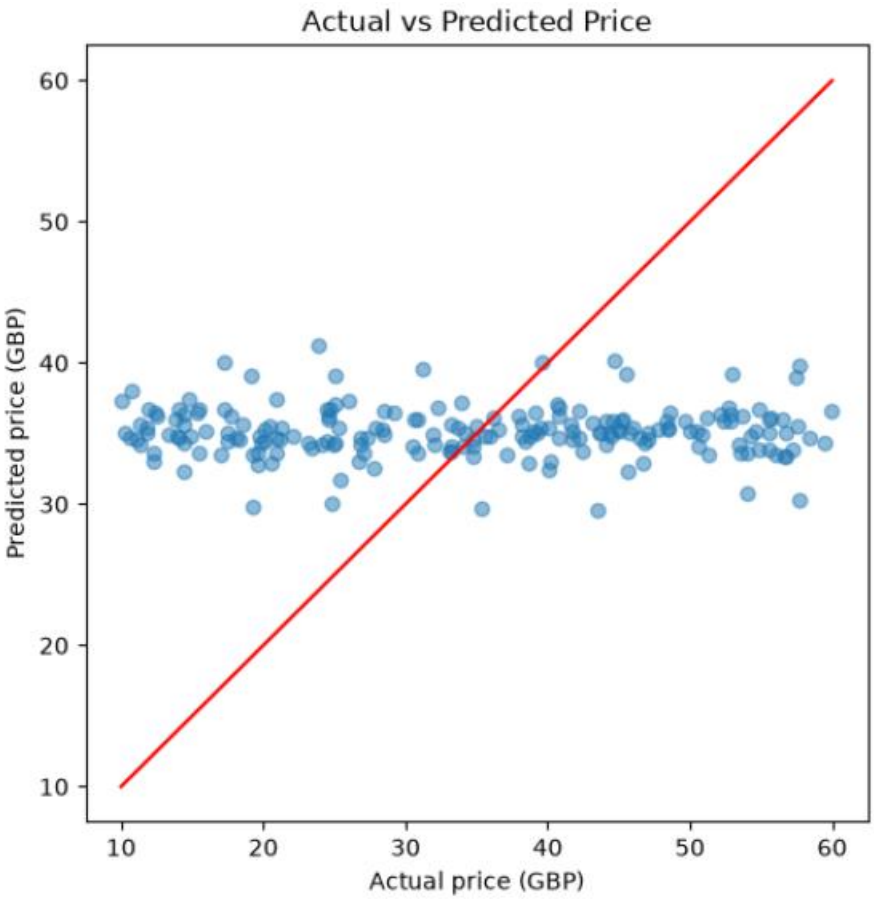


Figure 8: Actual versus predicted price on the test set

Figure 8 shows the predictions bunched in a narrow band around the average price, while the actual prices cover the whole range from £10 to £60. A good model would place the dots close to the red diagonal line, but here they do not.

1. Model limitations

- Books to Scrape is a practice website, so its prices are not set by real market forces and appear to be close to random. This is the most likely reason the model failed to learn a pattern.
- Only a few simple features were used. Real prices also depend on author, publisher, format and edition, which are not available here.
- Linear Regression only captures straight-line relationships.
- Genres with few books give unreliable coefficients, and the “Default” and “Add a comment” labels are not true genres.
- The data is a single snapshot in time, so trends over time cannot be studied.

Practical Implications

The main practical message is that, on this catalogue, price is not explained by the publicly visible book information. Star rating, stock level, title length and description length show almost no link with price, so a customer cannot use price to judge quality, and a retailer cannot use these simple features to set prices.

There is a small genre effect, with Travel and Fantasy books costing about £8 more on average than Food and Drink or Thriller books. A retailer could use this as a starting point when planning margins by genre, but it is a modest difference and should be checked against real sales data before it is relied on. The failure of the model also shows the value of testing against a baseline: without the comparison, an average error of £12.69 could wrongly look acceptable.

Recommendations

1 For retailers

- Review pricing by genre, especially in higher-priced genres such as Travel and Fantasy and lower-priced ones such as Food and Drink, to check that the differences match customer demand.
- Do not set prices from star rating or stock level alone, since neither showed a relationship with price.
- Fix the “Default” and “Add a comment” category labels so that these books can be analysed by genre.

2 For publishers

- Collect richer information, such as author, format, page count and sales history, because the simple fields available here could not predict price.
- Improve listing information, because richer product descriptions support customer decisions.

3 For data-driven book buyers

- Compare prices within the same genre rather than across genres, because average prices differ by up to about £8.
- Do not rely on price as a sign of quality, since higher-rated books were not more expensive.

Conclusion

This project collected 1 000 books from Books to Scrape, cleaned the data, explored it visually and built a Linear Regression model to estimate price. The main finding is that price is spread evenly between £10 and £60 and is almost unrelated to star rating, stock quantity, title length or description length, with only a small genre effect. The model achieved an R-squared of -0.022 and an MAE of £12.69, which is no better than the baseline of £12.66, so it cannot predict price from this data. Future work could add more features such as author or publisher, try other models such as decision trees, and use data from a real bookstore to test whether these findings hold more widely.

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